

# Dual-Site Enterprise Network Design

By: Andres Jose L. Tadeo

# A Little About This Project

*Hi, I am Andres. I built this project to challenge myself and prepare for designing and configuring networks similar to those used in real-world enterprise environments.*

*My goal was to see whether I could build a secure, professional-grade network from scratch that connects two separate sites while implementing routing, VPNs, security policies, and essential network services.*

**Packet Tracer Version Used:** v9.0.0.0810

Download: <https://www.netacad.com/articles/news/download-cisco-packet-tracer>

# Project Objectives

## **Objectives**

- Hierarchical Enterprise Design
- Secure Site-to-Site Connectivity
- Dynamic Routing
- Public and Private Services
- Redundancy and High Availability
- Layered Security Architecture

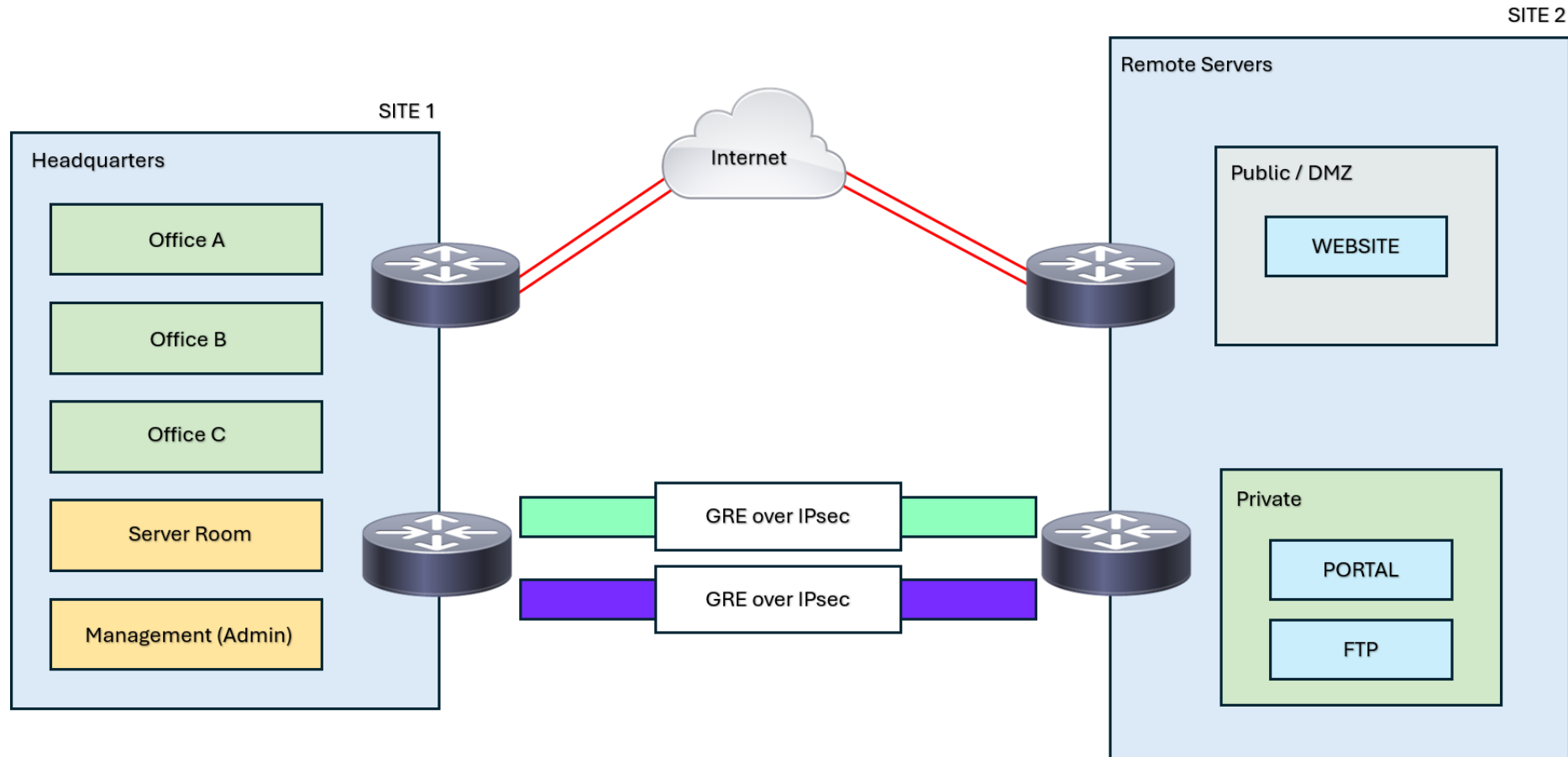
# Topology Overview

The network topology represents a dual-site enterprise WAN architecture connecting the Headquarters and Remote Site through redundant GRE over IPsec tunnels across WAN connections.

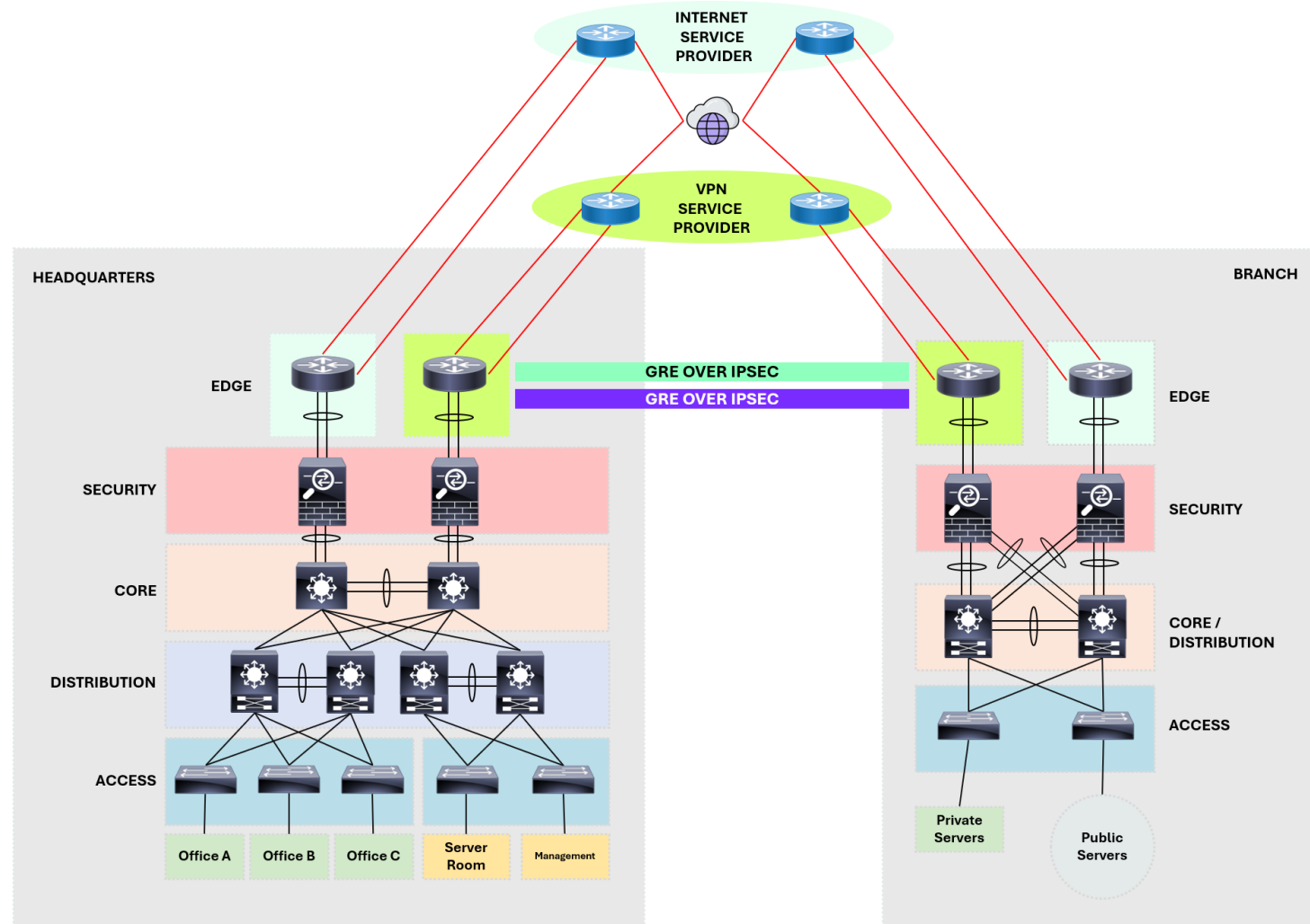
The design incorporates a three-tier campus architecture, VLAN segmentation, Layer 3 switching, dynamic routing, gateway redundancy, and perimeter firewalls to provide scalability, resiliency, and secure inter-site communication.

Dedicated edge routers separate Internet access and VPN functions, with one router handling Internet connectivity and the other managing GRE over IPsec tunnel operations.

# Simplified Topology View



# Full Topology View



# Headquarters

The Headquarters network uses a **Three-Tier Architecture** (Access, Distribution, and Core) to provide scalability, redundancy, and efficient traffic management.

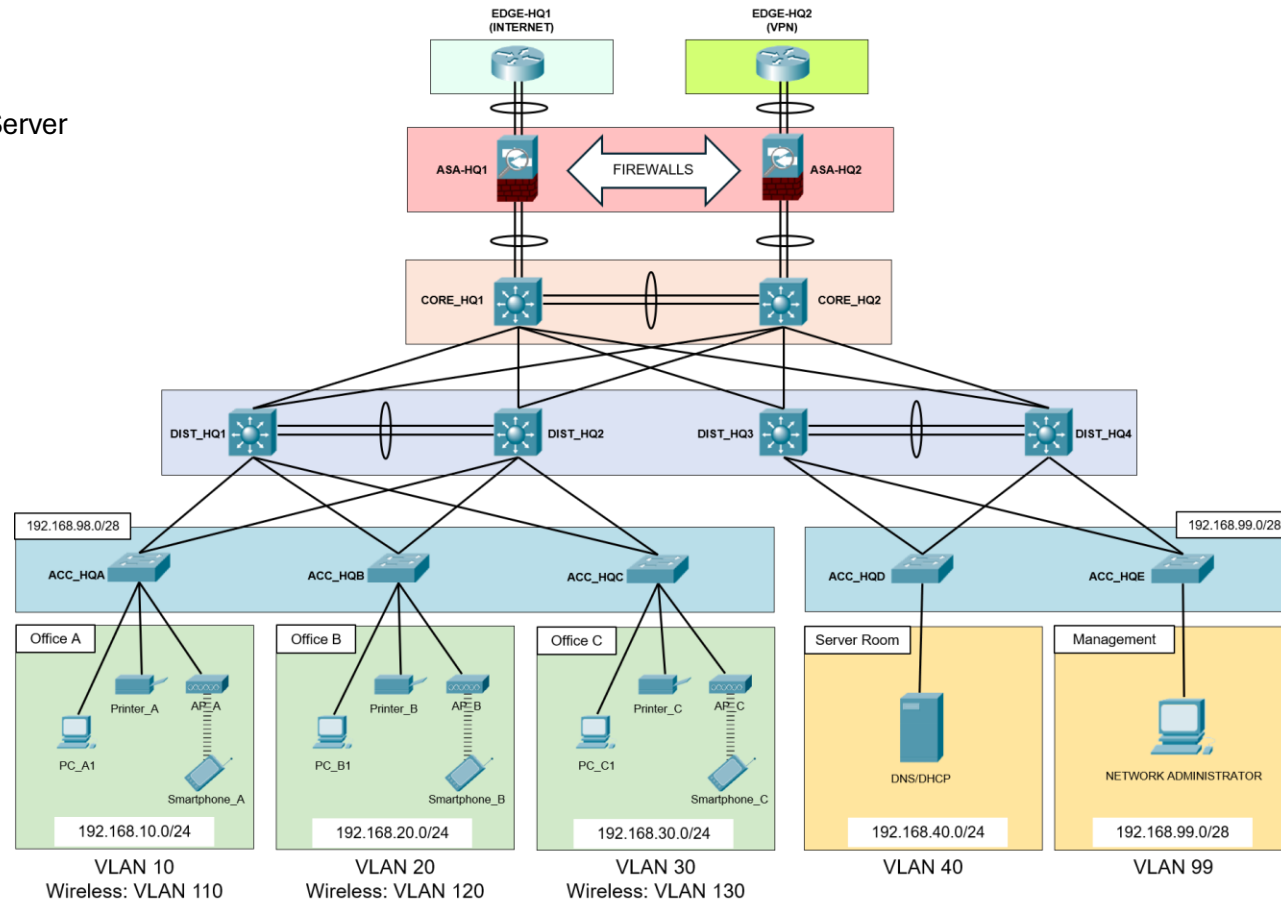
The network is segmented into Office A, Office B, Office C, Server Room, and Management VLANs, with a centralized server providing DHCP and DNS services.

## Components

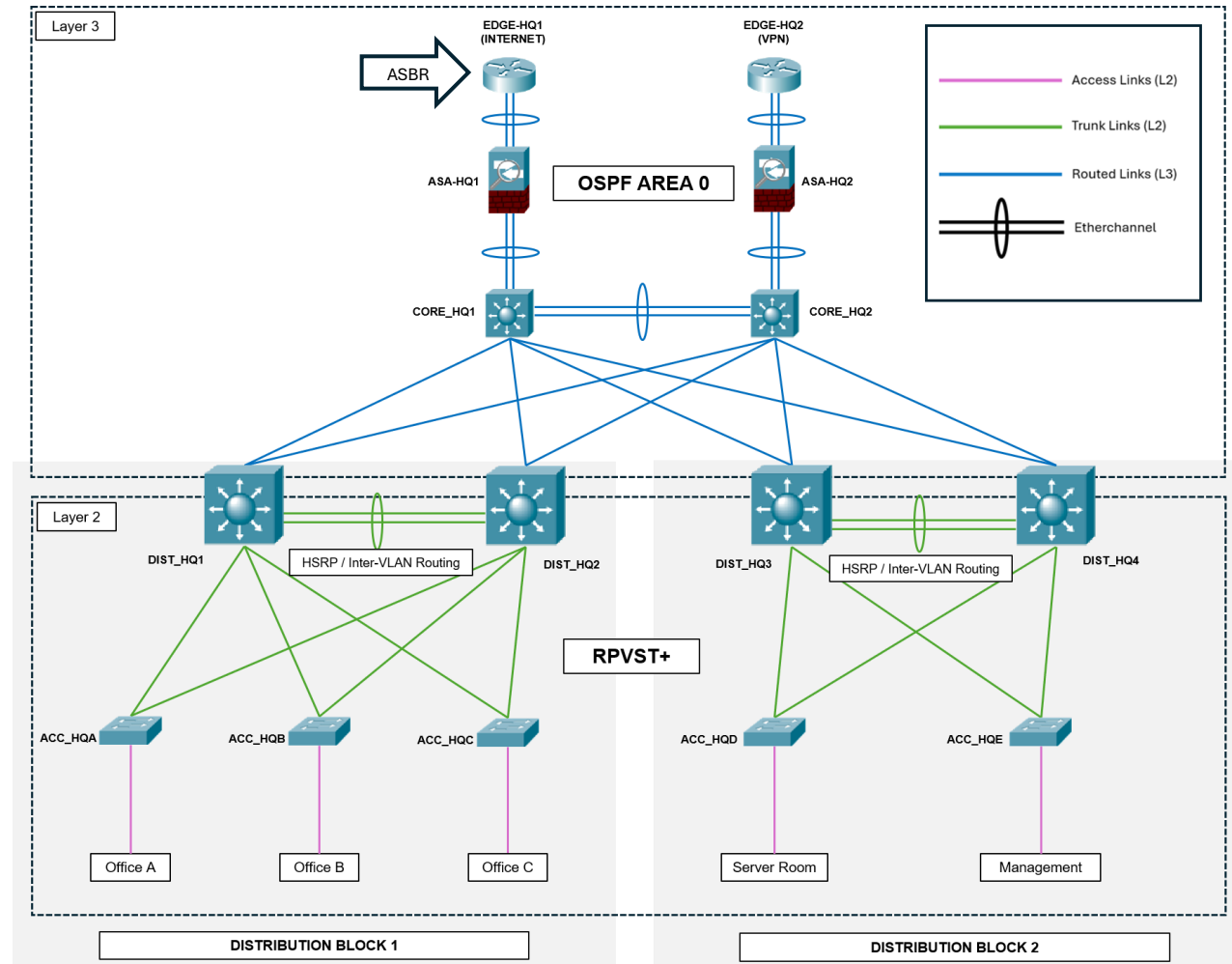
- Access Switches
- Distribution Switches
- Core Switches
- Dual Firewalls
- Dual Edge Routers

## Design Features

- Layered architecture
- EtherChannels
- OSPF routing
- Redundant paths
- HSRP
- RPVST+



# Headquarters Network Architecture





# Branch (Remote Servers)

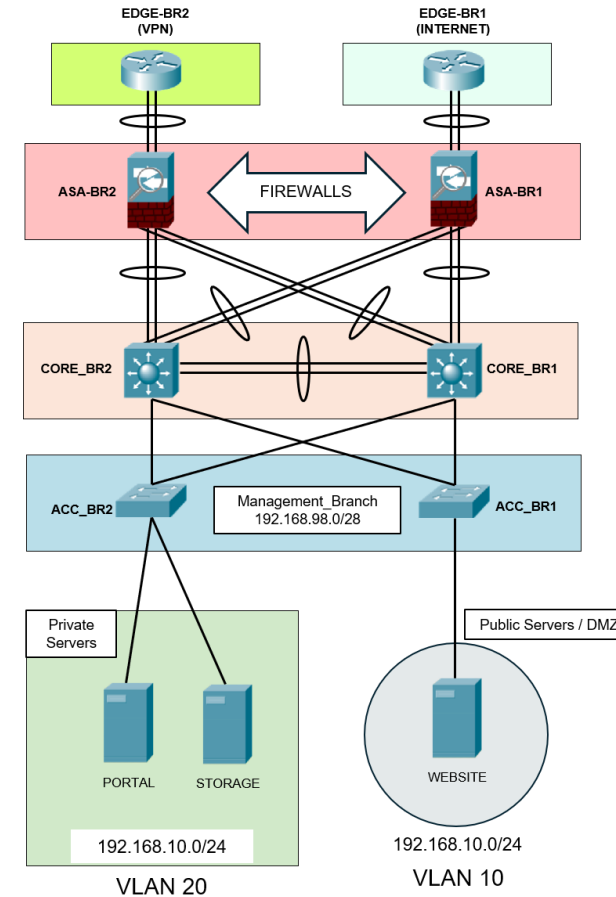
The **Remote Servers** network is designed to securely host services by separating public-facing and internal resources. It uses a **Two-tier Collapsed Core Architecture**, with core and distribution functions combined on the CORE\_BR1 and CORE\_BR2 switches.

## Components

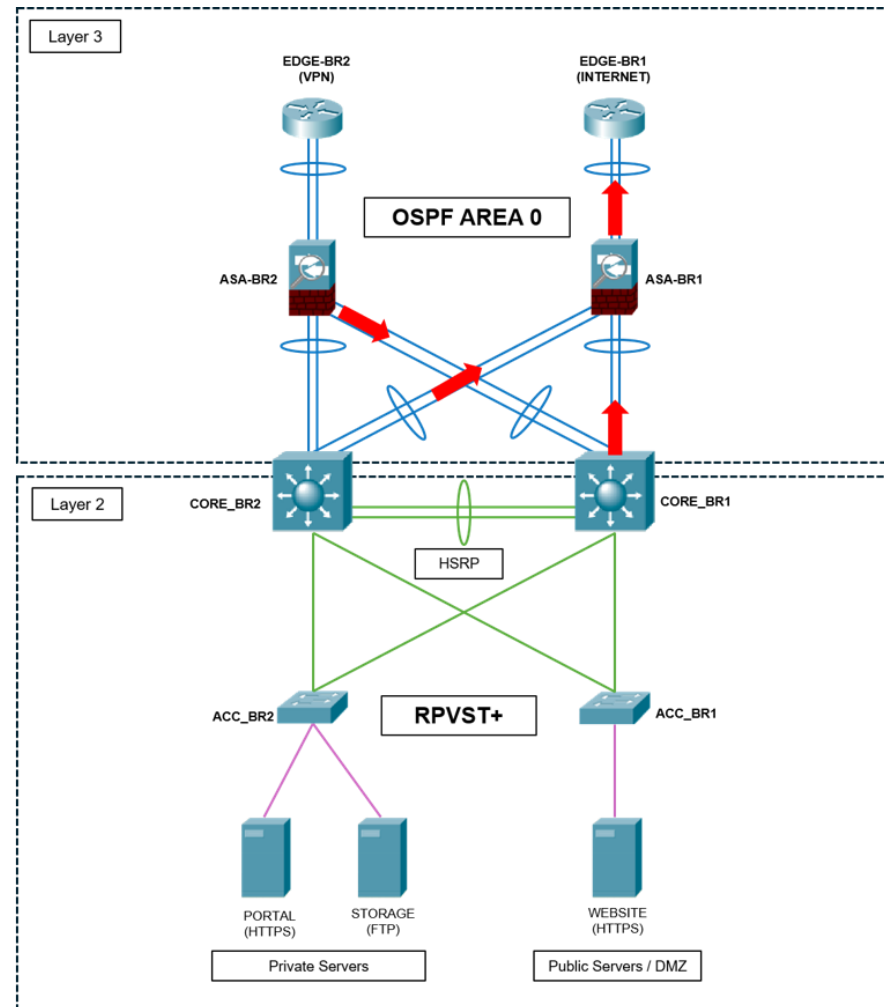
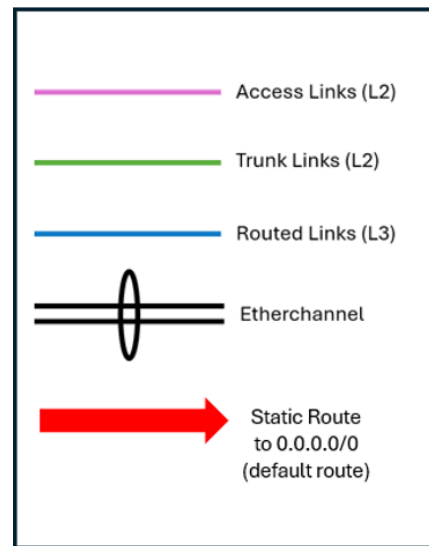
- Core Switches
- Access Switches
- Internal Server Segment
- DMZ Network
- Dual Firewalls
- Dual Edge Routers

## Design Features

- Secure server hosting
- DMZ isolation
- Layered architecture
- EtherChannels
- OSPF routing
- RIPv2+

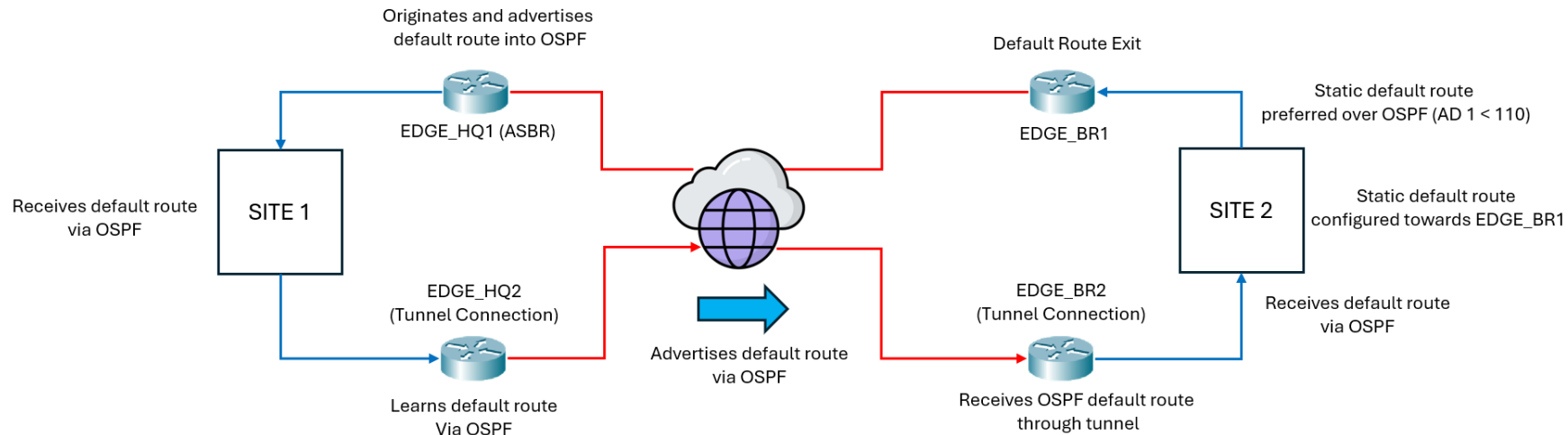


# Branch Network Architecture



# Why use Static Default Route?

- Packet Tracer supports only a limited set of Cisco IOS features and does not fully support advanced OSPF default-route behavior
- The `default-information originate <metric-type>` command is not available in Packet Tracer.
- Policy-Based Routing (PBR) is also not supported.
- A static default route has an Administrative Distance of 1, making it more preferred than an OSPF route with an Administrative Distance of 110.
- The result is predictable route selection while preserving OSPF route propagation across both sites.



# VLAN Architecture

| HEADQUARTERS |                   |                  |               |             |              |
|--------------|-------------------|------------------|---------------|-------------|--------------|
| VLAN         | Name              | Subnet           | Gateway (VIP) | HSRP Active | HSRP Standby |
| 10           | Office A          | 192.168.10.0/24  | 192.168.10.1  | DIST_HQ1    | DIST_HQ2     |
| 20           | Office B          | 192.168.20.0/24  | 192.168.20.1  | DIST_HQ1    | DIST_HQ2     |
| 30           | Office C          | 192.168.30.0/24  | 192.168.30.1  | DIST_HQ1    | DIST_HQ2     |
| 40           | Server_Room       | 192.168.40.0/24  | 192.168.40.1  | DIST_HQ3    | DIST_HQ4     |
| 99           | Management_12     | 192.168.98.0/28  | 192.168.98.1  | DIST_HQ2    | DIST_HQ1     |
|              | Management_34     | 192.168.99.0/28  | 192.168.99.1  | DIST_HQ4    | DIST_HQ3     |
| 110          | AP_A              | 192.168.110.0/24 | 192.168.110.1 | DIST_HQ2    | DIST_HQ1     |
| 120          | AP_B              | 192.168.120.0/24 | 192.168.120.1 | DIST_HQ2    | DIST_HQ1     |
| 130          | AP_C              | 192.168.130.0/24 | 192.168.130.1 | DIST_HQ2    | DIST_HQ1     |
| 1001         | Native VLAN       | X                | X             | X           | X            |
| BRANCH       |                   |                  |               |             |              |
| VLAN         | Name              | Subnet           | Gateway (VIP) | HSRP Active | HSRP Standby |
| 10           | Public_Servers    | 172.16.10.0/28   | 172.16.10.1   | DIST_HQ1    | DIST_HQ2     |
| 20           | Private_Servers   | 172.16.20.0/28   | 172.16.20.1   | DIST_HQ2    | DIST_HQ1     |
| 99           | Management_Branch | 172.16.99.0/28   | 172.16.99.1   | DIST_HQ2    | DIST_HQ1     |
| 1001         | Native VLAN       | X                | X             | X           | X            |

# Layer 2 Architecture

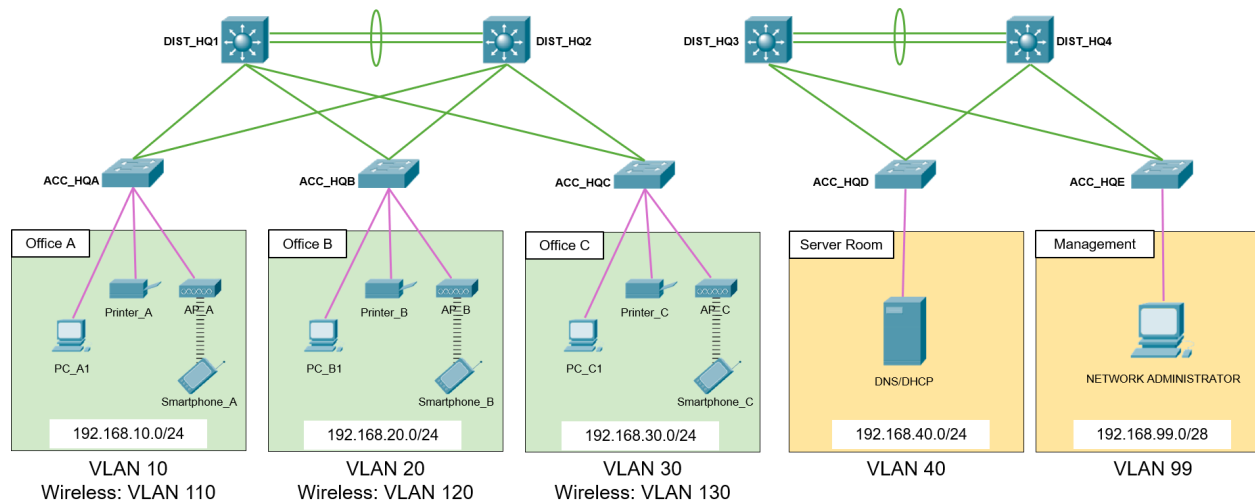
## Technologies

- EtherChannel
- Rapid PVST+
- PortFast
- BPDU Guard

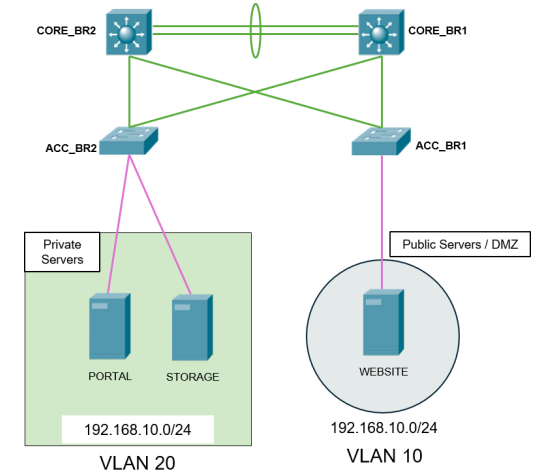
## Benefits

- Redundancy
- Fast convergence
- Loop prevention

## HEADQUARTERS



## Branch



# Spanning Tree Protocol Design (HQ)

**Rapid Per-VLAN Spanning Tree Plus (RPVST+)** is deployed across the switching infrastructure to provide fast Layer 2 convergence, prevent switching loops, and support redundant network paths.

Root Bridge election is aligned with HSRP active gateway assignments to ensure optimal traffic flow

## Additional STP Features:

### ➤ **PortFast**

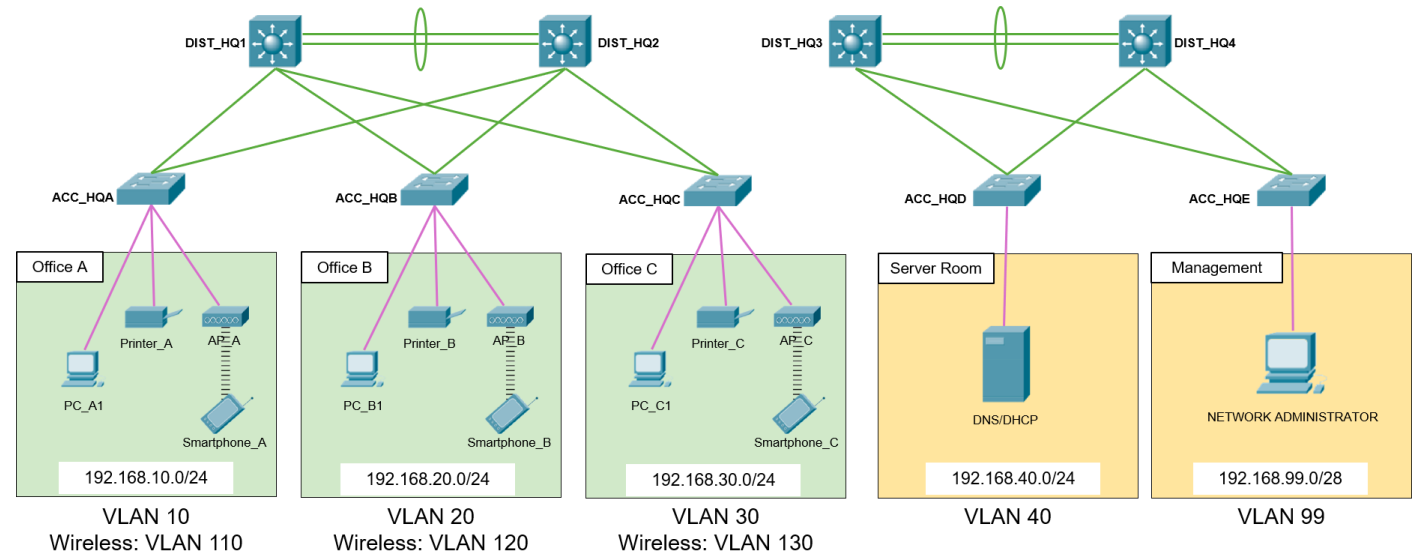
Configured on all access ports connected to:

- PCs
- Printers
- Servers
- End-user devices

### ➤ **BPDU Guard**

Enabled on all PortFast interfaces.

| ROOT BRIDGE PLACEMENT |              |                |
|-----------------------|--------------|----------------|
| VLAN                  | Primary Root | Secondary Root |
| 10, 20, 30            | DIST_HQ1     | DIST_HQ2       |
| 99, 110, 120, 130     | DIST_HQ2     | DIST_HQ1       |
| 40                    | DIST_HQ3     | DIST_HQ4       |
| 99                    | DIST_HQ4     | DIST_HQ3       |



# Spanning Tree Protocol Design (Branch)

**Rapid Per-VLAN Spanning Tree Plus (RPVST+)** is deployed across the switching infrastructure to provide fast Layer 2 convergence, prevent switching loops, and support redundant network paths.

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## Additional STP Features:

### ➤ **PortFast**

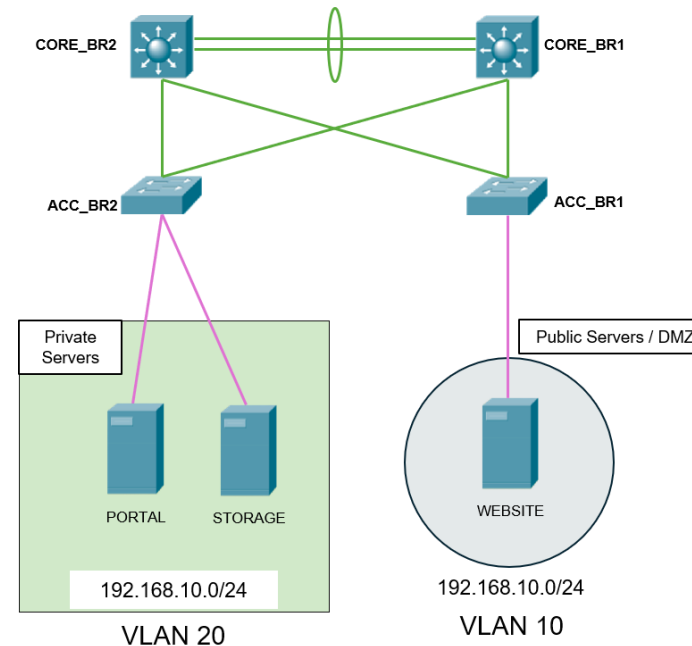
Configured on all access ports connected to:

- Private Servers
- Public Servers

### ➤ **BPDU Guard**

Enabled on all PortFast interfaces.

| ROOT BRIDGE PLACEMENT |              |                |
|-----------------------|--------------|----------------|
| VLAN                  | Primary Root | Secondary Root |
| 10                    | CORE_BR1     | CORE_BR2       |
| 20, 99                | CORE_BR2     | CORE_BR1       |

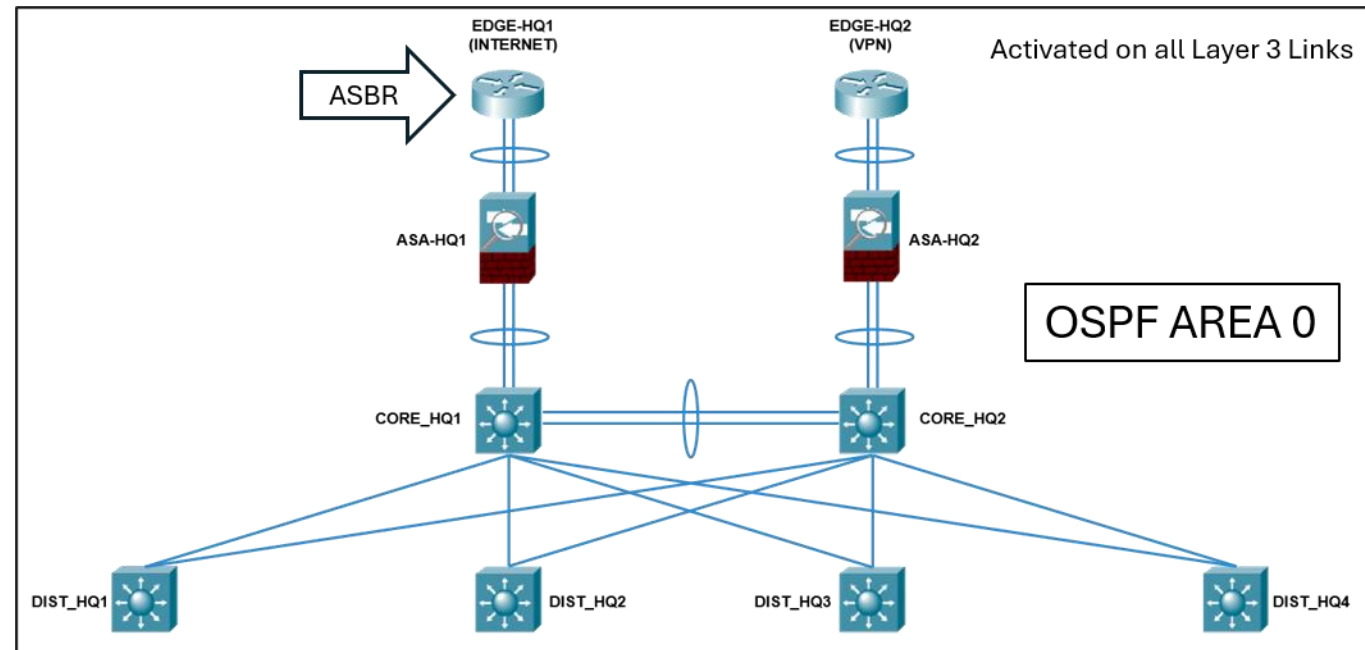


# OSPF Design (HQ)

**Open Shortest Path First** (Single Area) is selected as the dynamic routing protocol to provide efficient connectivity between the Layer 3 Switches, Firewalls and Edge Routers, it automatically calculates the best paths for traffic without manual route configuration.

To maintain stability, VLAN and Loopback interfaces are configured as `passive-interface` to prevent unnecessary adjacencies, while primary and backup default routes (0.0.0.0/0) are injected at EDGE\_HQ1 (**Autonomous System Boundary Router**) to manage all outbound traffic.

| Device   | Router-ID     |
|----------|---------------|
| DIST_HQ1 | 10.255.255.1  |
| DIST_HQ2 | 10.255.255.2  |
| DIST_HQ3 | 10.255.255.3  |
| DIST_HQ4 | 10.255.255.4  |
| CORE_HQ1 | 10.255.255.10 |
| CORE_HQ2 | 10.255.255.11 |
| ASA-HQ1  | 10.255.255.15 |
| ASA-HQ2  | 10.255.255.16 |
| EDGE_HQ1 | 10.255.255.21 |
| EDGE_HQ2 | 10.255.255.22 |





# Routing Verification (HQ)

This is a sample routing table from **CORE\_HQ1** that shows **OSPF** has successfully exchanged and installed routes across the network, providing reachability to all required destinations.

Multiple equal-cost paths are available for several networks, enabling **ECMP** (Equal-Cost Multi-Path) to improve overall network performance and availability.

## ECMP provides:

- Redundancy
- Load Balancing
- Scalability
- Failover
- Efficiency
- Reliability

```
CORE_HQ1#show ip route ospf
 10.0.0.0/8 is variably subnetted, 27 subnets, 2 masks
O    10.10.20.0 [110/2] via 10.10.10.1, 00:00:37, GigabitEthernet1/0/1
    [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.10.40.0 [110/2] via 10.10.30.1, 00:00:37, GigabitEthernet1/0/2
    [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.10.60.0 [110/2] via 10.10.50.1, 00:00:37, GigabitEthernet1/0/3
    [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.10.80.0 [110/2] via 10.10.70.1, 00:00:37, GigabitEthernet1/0/4
    [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.20.10.4 [110/2] via 10.20.10.2, 00:00:37, Port-channel1
O    10.20.20.0 [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.20.20.4 [110/3] via 10.5.5.2, 00:00:37, Port-channel2
O    10.255.255.1 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
O    10.255.255.2 [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O    10.255.255.3 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
O    10.255.255.4 [110/2] via 10.10.70.1, 00:01:07, GigabitEthernet1/0/4
O    10.255.255.12 [110/2] via 10.5.5.2, 00:00:37, Port-channel2
O    10.255.255.21 [110/3] via 10.20.10.2, 00:00:37, Port-channel1
O    10.255.255.22 [110/4] via 10.5.5.2, 00:00:37, Port-channel2
O   192.168.10.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O   192.168.20.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O   192.168.30.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O   192.168.40.0 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
    [110/2] via 10.10.70.1, 00:01:07, GigabitEthernet1/0/4
192.168.98.0/28 is subnetted, 1 subnets
O   192.168.98.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
192.168.99.0/28 is subnetted, 1 subnets
O   192.168.99.0 [110/2] via 10.10.50.1, 00:01:07, GigabitEthernet1/0/3
O   192.168.110.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O   192.168.120.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O   192.168.130.0 [110/2] via 10.10.10.1, 00:01:07, GigabitEthernet1/0/1
    [110/2] via 10.10.30.1, 00:01:07, GigabitEthernet1/0/2
O*E2 0.0.0.0/0 [110/1] via 10.20.10.2, 00:00:37, Port-channel1
```

Default Route learned from ASBR



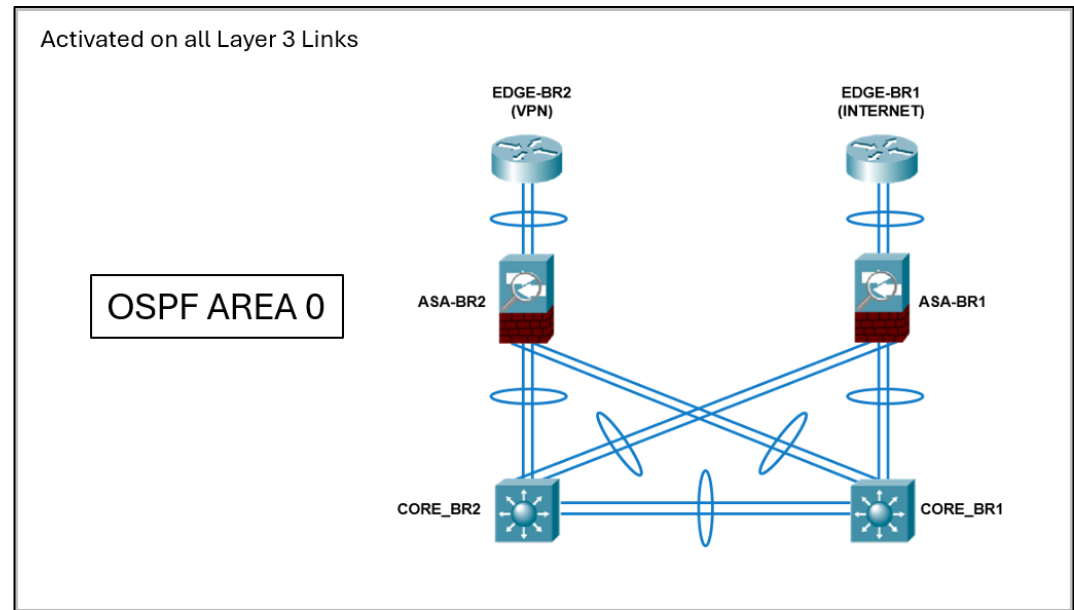
O\*E2 0.0.0.0/0 [110/1] via 10.20.10.2, 00:00:37, Port-channel1

# OSPF Design (Branch)

**Open Shortest Path First** (Single Area) is selected as the dynamic routing protocol to provide efficient connectivity between the Layer 3 Switches, Firewalls and Edge Routers, it automatically calculates the best paths for traffic without manual route configuration.

To maintain stability, VLAN and Loopback interfaces are configured as `passive-interface` to prevent unnecessary adjacencies

| Device   | Router-ID     |
|----------|---------------|
| CORE_BR1 | 10.255.255.31 |
| CORE_BR1 | 10.255.255.32 |
| ASA-BR1  | 10.255.255.35 |
| ASA-BR2  | 10.255.255.36 |
| EDGE_BR1 | 10.255.255.41 |
| EDGE_BR2 | 10.255.255.42 |



# Routing Verification (Branch)

This is a sample routing table from CORE\_BR1 that shows OSPF has successfully exchanged and installed routes across the network, providing reachability to all required destinations.

A static route is configured towards the external router was configured.

Static default route selected because AD 1 is lower than OSPF AD 110.

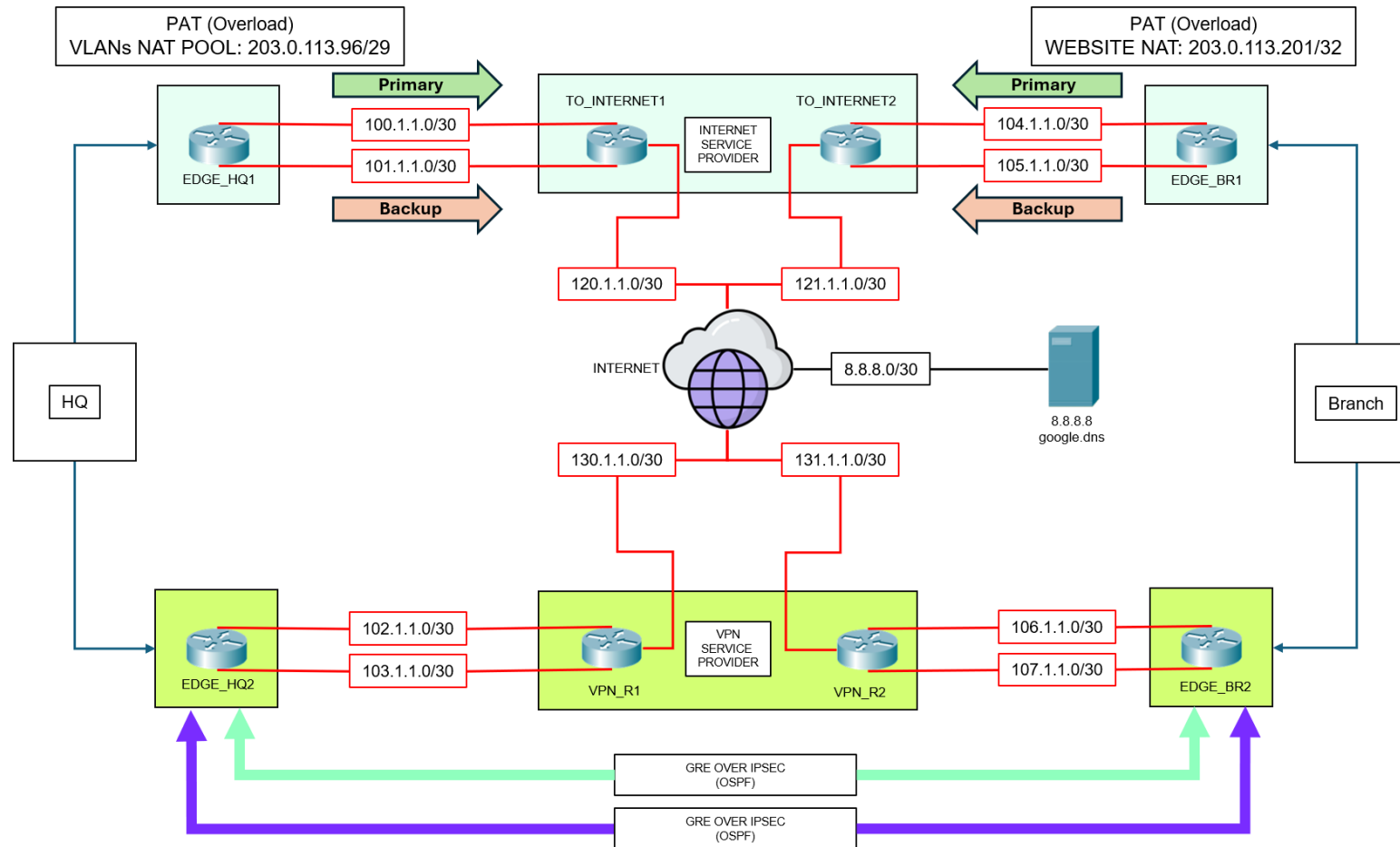


```
CORE_BR1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.30.10.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.30.10.0/30 is directly connected, Port-channel1
L       10.30.10.1/32 is directly connected, Port-channel1
O       10.30.10.4/30 [110/2] via 10.30.10.2, 00:01:12, Port-channel1
O       10.30.10.8/30 [110/51] via 172.16.99.3, 00:01:12, Vlan99
        [110/51] via 10.30.10.2, 00:01:12, Port-channel1
O       10.30.20.0/30 [110/2] via 172.16.99.3, 00:01:12, Vlan99
O       10.30.20.4/30 [110/3] via 172.16.99.3, 00:01:02, Vlan99
C       10.30.20.8/30 is directly connected, Port-channel3
L       10.30.20.9/32 is directly connected, Port-channel3
C       10.255.255.31/32 is directly connected, Loopback0
O       10.255.255.32/32 [110/2] via 172.16.99.3, 00:01:12, Vlan99
O       10.255.255.41/32 [110/3] via 10.30.10.2, 00:01:02, Port-channel1
O       10.255.255.42/32 [110/4] via 172.16.99.3, 00:01:02, Vlan99
    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.10.0/28 is directly connected, Vlan10
L       172.16.10.2/32 is directly connected, Vlan10
C       172.16.20.0/28 is directly connected, Vlan20
L       172.16.20.2/32 is directly connected, Vlan20
C       172.16.99.0/28 is directly connected, Vlan99
L       172.16.99.2/32 is directly connected, Vlan99
S*    0.0.0.0/0 [1/0] via 10.30.10.2
```

# WAN Architecture



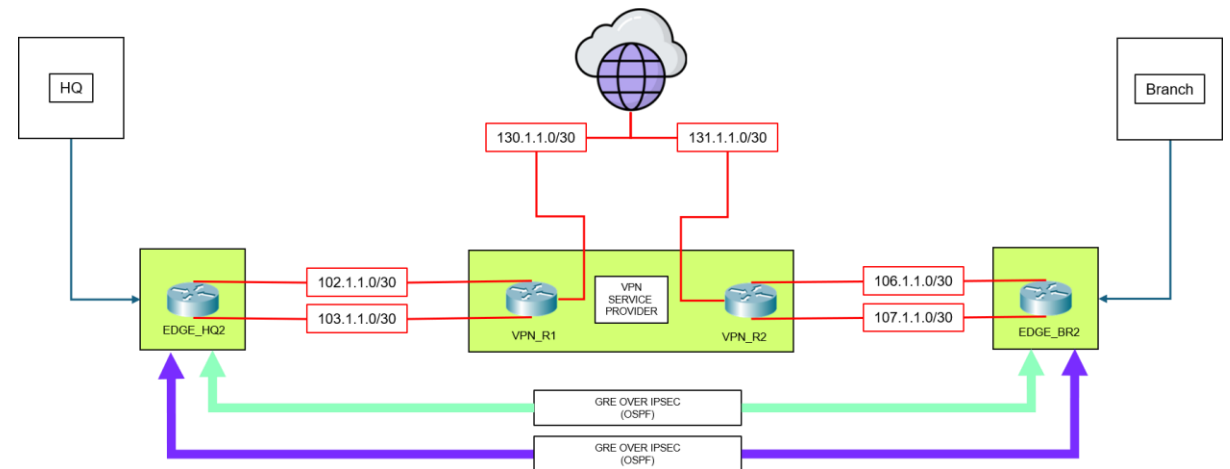
# WAN Routing Summary

| Path                    | Network      | Connected Devices              | Purpose                                                                     | Routing                                                                 |
|-------------------------|--------------|--------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------|
| HQ Internet Primary     | 100.1.1.0/30 | EDGE_HQ1 ↔ TO_INTERNET1        | Internet Access<br>(Public NAT prefixes advertised via static routes)       | Static Route (Primary)                                                  |
| HQ Internet Backup      | 101.1.1.0/30 |                                |                                                                             | Floating Static Route                                                   |
| Branch Internet Primary | 104.1.1.0/30 | EDGE_BR1 ↔ TO_INTERNET2        |                                                                             | Static Route (Primary)                                                  |
| Branch Internet Backup  | 105.1.1.0/30 |                                |                                                                             | Floating Static Route                                                   |
| HQ VPN                  | 102.1.1.0/30 | EDGE_HQ2 ↔ VPN_R1              | GRE/IPsec Transport                                                         | Static Route<br>(Inter-site routing is handled by GRE Tunnels via OSPF) |
|                         | 103.1.1.0/30 |                                |                                                                             |                                                                         |
| Branch VPN              | 106.1.1.0/30 | EDGE_BR2 ↔ VPN_R2              |                                                                             |                                                                         |
|                         | 107.1.1.0/30 |                                |                                                                             |                                                                         |
| ISP Transit             | 120.1.1.0/30 | ISP Routers ↔ Internet Routers | Internet Connectivity<br>(Public NAT prefixes advertised via static routes) | Static Routes                                                           |
|                         | 121.1.1.0/30 |                                |                                                                             |                                                                         |
| VPN Transit             | 130.1.1.0/30 | VPN Routers ↔ Internet Routers | Tunnel Transport                                                            | Static Routes                                                           |
|                         | 131.1.1.0/30 |                                |                                                                             |                                                                         |
| Public Service          | 8.8.8.0/30   | INTERNET ↔ google.dns          | Public Service Access                                                       | Static Routes                                                           |

# GRE Tunneling

**GRE Tunneling** was implemented to provide a logical point-to-point connection between sites and to support OSPF route exchange, enabling dynamic routing across the WAN. Security is provided separately through **IPsec** encryption.

| TUNNEL OVERLAY TRANSPORT  |                |                    |                  |                  |
|---------------------------|----------------|--------------------|------------------|------------------|
| Tunnel                    | Network        | HQ Tunnel IP       | Branch Tunnel IP | Routing Protocol |
| Tunnel 1                  | 10.0.1.0/30    | 10.0.1.1           | 10.0.1.2         | OSPF             |
| Tunnel 2                  | 10.0.2.0/30    | 10.0.2.1           | 10.0.2.2         | OSPF             |
| TUNNEL UNDERLAY TRANSPORT |                |                    |                  |                  |
| Tunnel                    | HQ WAN Network | Branch WAN Network | Transport        |                  |
| Tunnel 1                  | 102.1.1.0/30   | 106.1.1.0/30       | GRE over IPsec   |                  |
| Tunnel 2                  | 103.1.1.0/30   | 107.1.1.0/30       | GRE over IPsec   |                  |



# IPsec Parameters

| IPSEC PHASE 1 (IKE/ISAKMP) CONFIGURATION |                |                    |            |                |          |                |
|------------------------------------------|----------------|--------------------|------------|----------------|----------|----------------|
| Tunnel                                   | Local Peer     | Remote Peer        | Encryption | Authentication | DH Group | Pre-Shared Key |
| Tunnel 1                                 | 102.1.1.1 (HQ) | 106.1.1.1 (Branch) | AES-128    | Pre-Shared Key | Group 5  | AJ123          |
| Tunnel 2                                 | 103.1.1.1 (HQ) | 107.1.1.1 (Branch) |            |                |          | AJ123          |

| IPSEC PHASE 2 (IPSEC SA) CONFIGURATION |                             |               |            |              |            |  |
|----------------------------------------|-----------------------------|---------------|------------|--------------|------------|--|
| Tunnel                                 | Interesting Traffic (ACL)   | Transform Set | Encryption | Integrity    | Crypto Map |  |
| Tunnel 1                               | (GRE) 102.1.1.1 ↔ 106.1.1.1 | T1-SET        | ESP-AES    | ESP-SHA-HMAC | T1-MAP     |  |
| Tunnel 2                               | (GRE) 103.1.1.1 ↔ 107.1.1.1 | T2-SET        |            |              | T2-MAP     |  |

| CRYPTO MAP APPLICATION |              |                  |
|------------------------|--------------|------------------|
| Tunnel                 | HQ Interface | Branch Interface |
| Tunnel 1               | G0/1/0       | G0/1/0           |
| Tunnel 2               | G0/2/0       | G0/2/0           |

# SERVICES – DHCP & DNS (HQ)

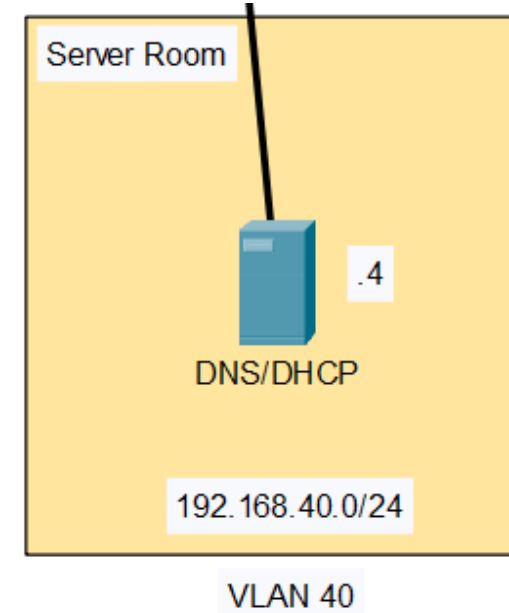
A combined **DHCP** and **DNS** server is deployed in HQ to centralize services. This setup simplifies administration and ensures consistent and convenient IP management across all offices.

DHCP Pool Table

| Pool Name | Default Gateway | DNS Server   | Start IP Address | Subnet Mask   | Max User |
|-----------|-----------------|--------------|------------------|---------------|----------|
| Office_A  | 192.168.10.1    | 192.168.40.4 | 192.168.10.11    | 255.255.255.0 | 245      |
| Office_B  | 192.168.20.1    | 192.168.40.4 | 192.168.20.11    | 255.255.255.0 | 245      |
| Office_C  | 192.168.30.1    | 192.168.40.4 | 192.168.30.11    | 255.255.255.0 | 245      |
| AP_A      | 192.168.110.1   | 192.168.40.4 | 192.168.110.11   | 255.255.255.0 | 245      |
| AP_B      | 192.168.120.1   | 192.168.40.4 | 192.168.120.11   | 255.255.255.0 | 245      |
| AP_C      | 192.168.130.1   | 192.168.40.4 | 192.168.130.11   | 255.255.255.0 | 245      |

DNS Records Table

| No. | Name        | Type     | Detail        |
|-----|-------------|----------|---------------|
| 0   | ajtadeo.com | A Record | 203.0.113.201 |
| 1   | dns.google  | A Record | 8.8.8.8       |
| 2   | portal.hq   | A Record | 172.16.10.8   |



The **SVIs** on DIST\_HQ1 and DIST\_HQ2 are configured with `ip helper-address 192.168.40.4` command, allowing them to function as **DHCP Relay Agents**.

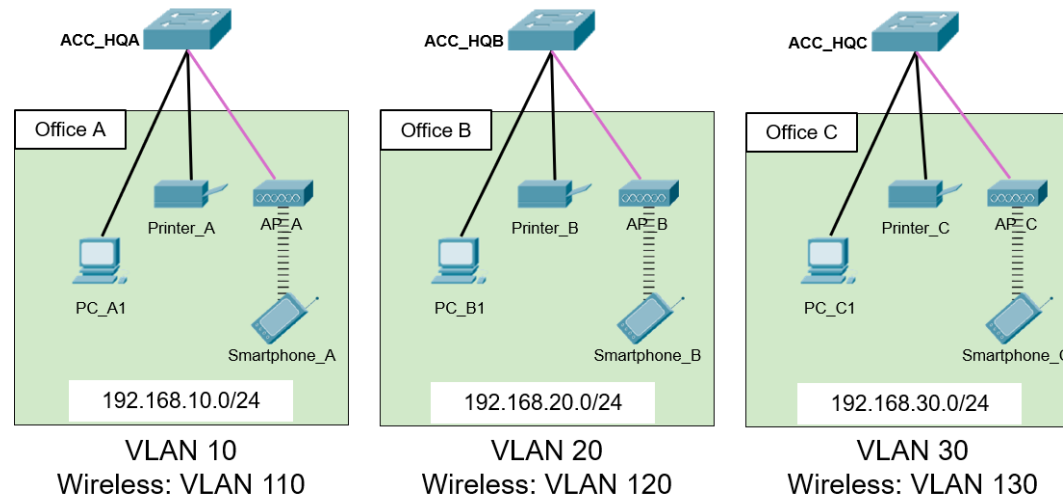
These agents listen for broadcast requests on end-user VLANs and send them to the central server as unicast packets.



# SERVICES – Wireless Connectivity (HQ)

The offices use AP\_PT-N access points operating on the IEEE 802.11n standard, supporting both **2.4 GHz** and **5 GHz** bands. Although IEEE 802.11ac provides higher performance, 802.11n was selected to ensure compatibility with devices that support only the 2.4 GHz band.

| Access Point | VLAN | SSID     | Authentication | Passphrase  |
|--------------|------|----------|----------------|-------------|
| AP_A         | 110  | Office_A | WPA2-PSK       | password123 |
| AP_B         | 120  | Office_B | WPA2-PSK       | password123 |
| AP_C         | 130  | Office_C | WPA2-PSK       | password123 |



# SERVICES – FTP & HTTPS (Branch)

## Portal Server (HTTPS)

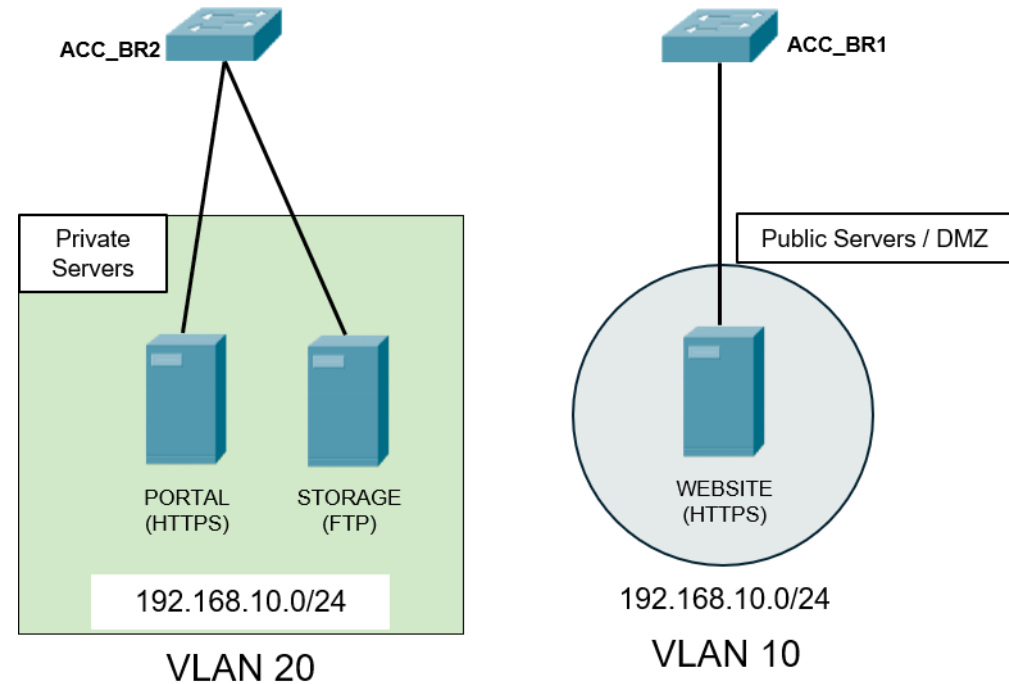
- Provides internal web services accessible only through the GRE over IPsec tunnels.
- The portal is intended for enterprise users and is not exposed to the Internet.

## Storage Server (FTP)

- Provides centralized file storage and transfer services for enterprise users.
- FTP access is restricted to internal networks and protected by firewall policies.

## Website Server (HTTPS)

- Hosts the organization's public-facing website.
- External users access the service through the translated public address (**203.0.113.201**) while the private address remains hidden.
- Direct access to the private IP is restricted to the Network Administrator.



# SERVICES – FTP & HTTPS (Branch)

## Portal Server (HTTPS)

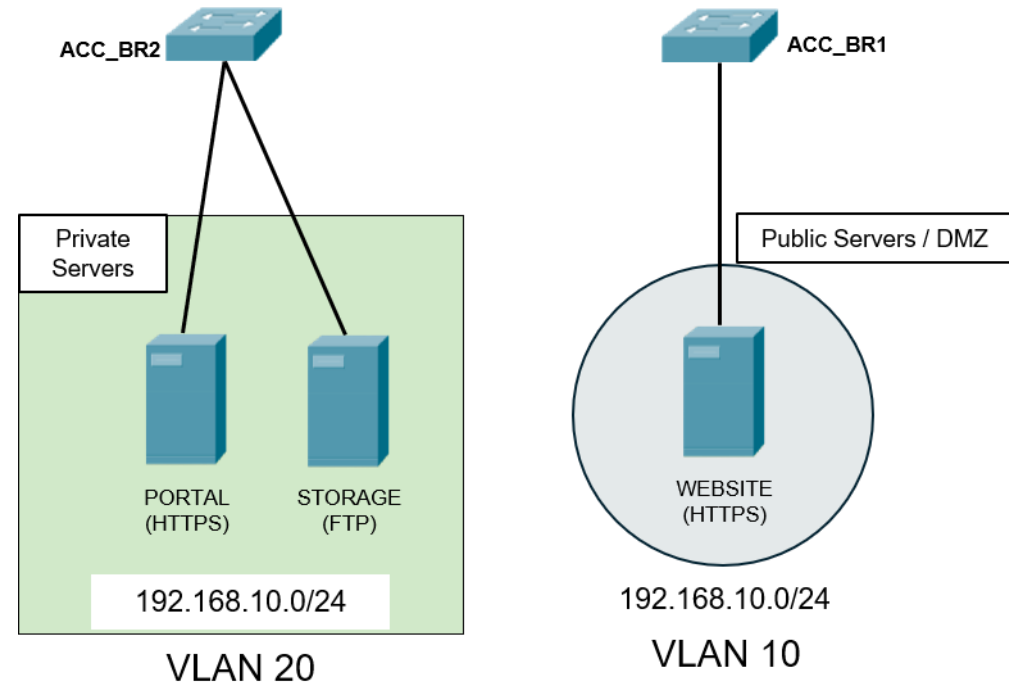
- Provides internal web services accessible only through the GRE over IPsec tunnels.
- The portal is intended for enterprise users and is not exposed to the Internet.

## Storage Server (FTP)

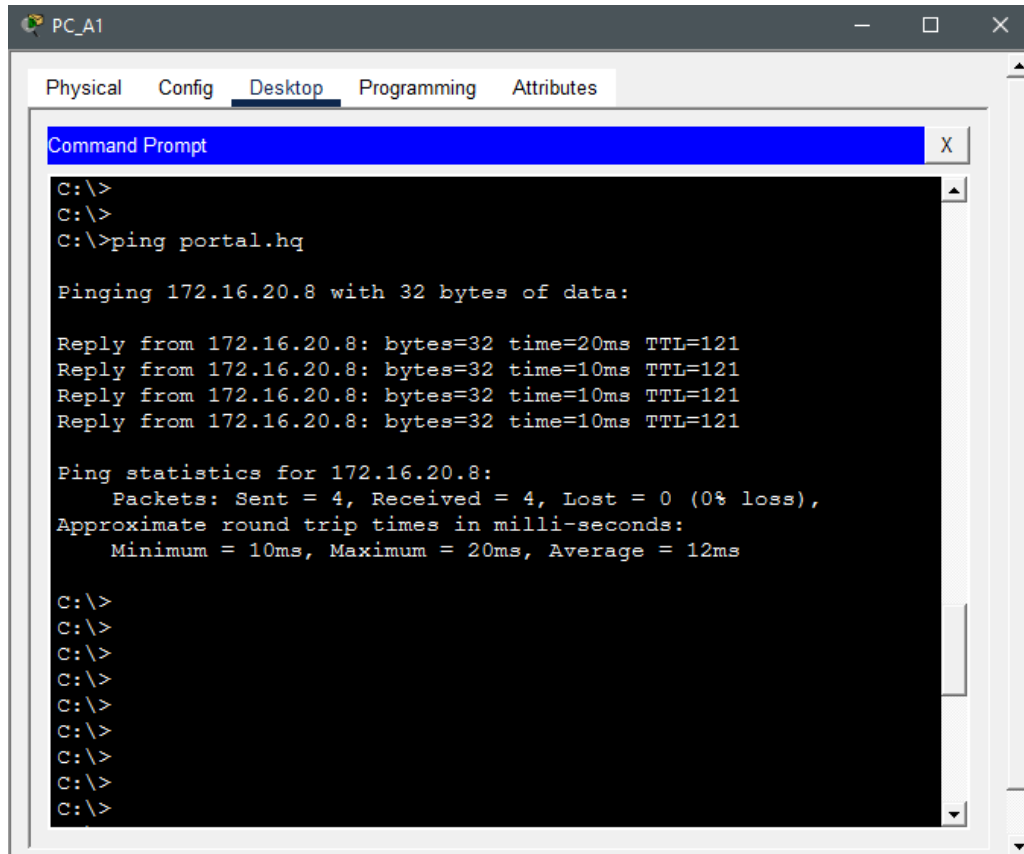
- Provides centralized file storage and transfer services for enterprise users.
- FTP access is restricted to internal networks and protected by firewall policies.

## Website Server (HTTPS)

- Hosts the organization's public-facing website.
- External users access the service through the translated public address (**203.0.113.201**) while the private address remains hidden.
- Direct access to the private IP is restricted to the Network Administrator.



# Verification – PORTAL(HTTPS)



The screenshot shows a PC\_A1 Desktop window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the following text:

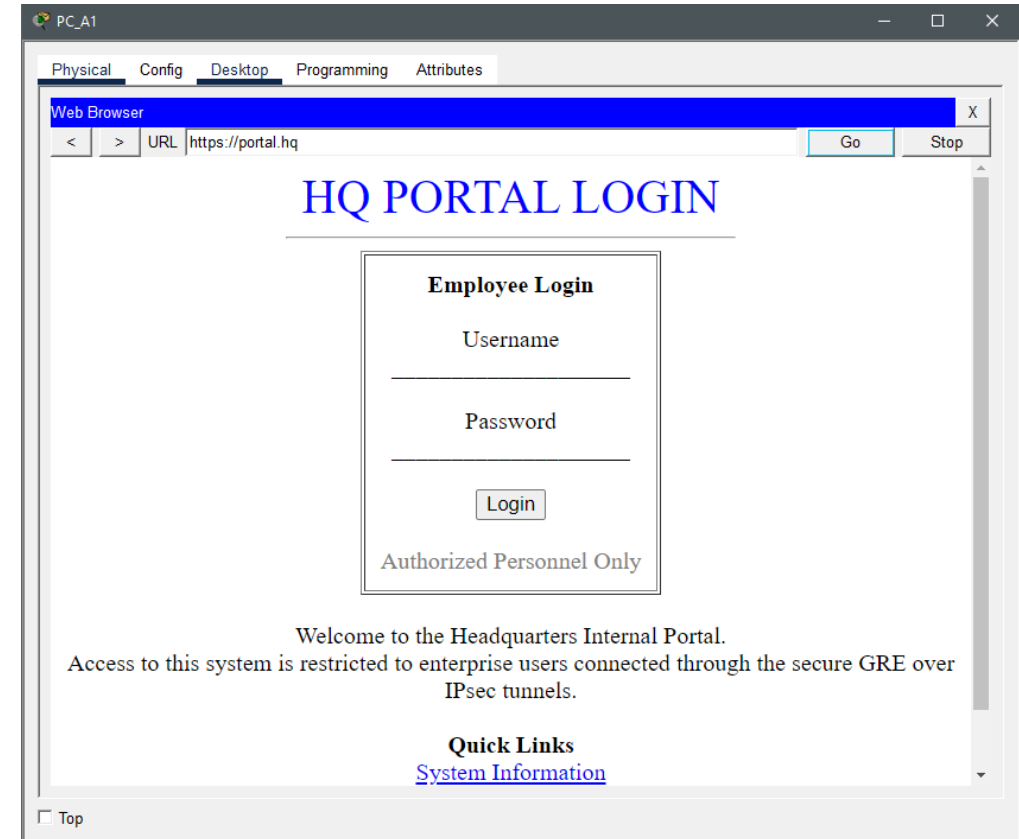
```
C:\>
C:\>
C:\>ping portal.hq

Pinging 172.16.20.8 with 32 bytes of data:

Reply from 172.16.20.8: bytes=32 time=20ms TTL=121
Reply from 172.16.20.8: bytes=32 time=10ms TTL=121
Reply from 172.16.20.8: bytes=32 time=10ms TTL=121
Reply from 172.16.20.8: bytes=32 time=10ms TTL=121

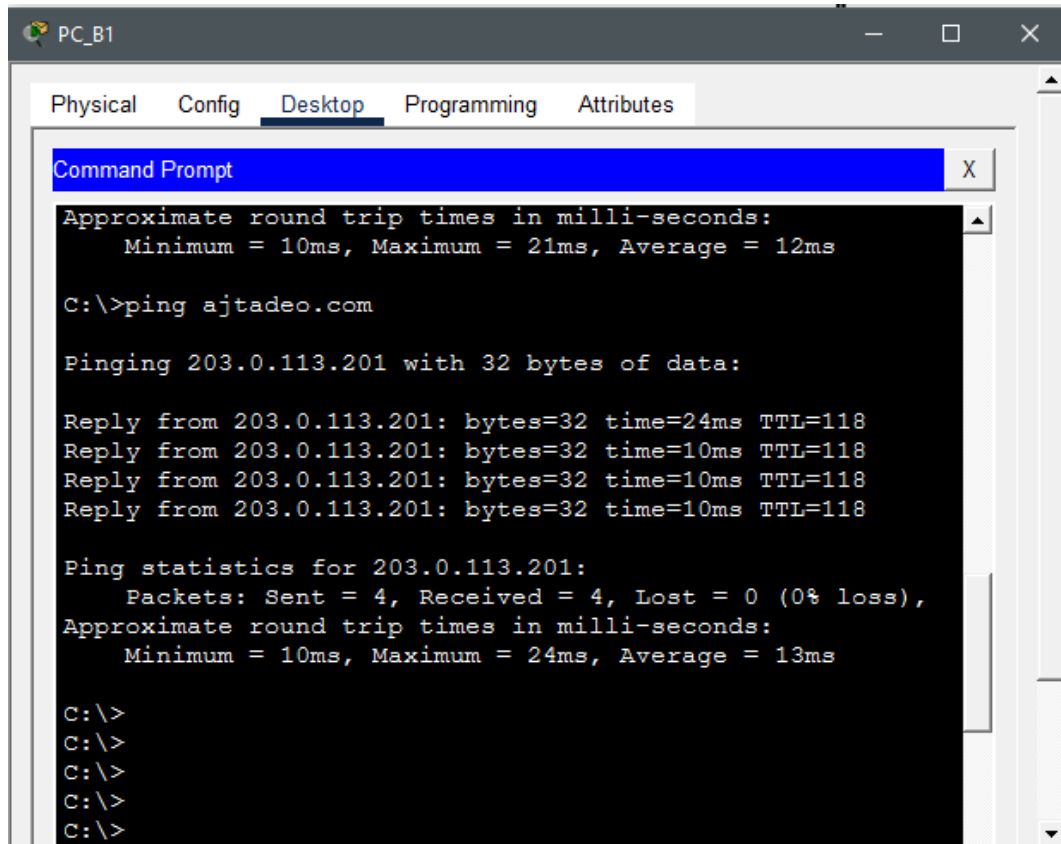
Ping statistics for 172.16.20.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 20ms, Average = 12ms

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```



*Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.*

# Verification – WEBSITE (HTTPS)



```
PC_B1
Physical Config Desktop Programming Attributes
Command Prompt
Approximate round trip times in milli-seconds:
  Minimum = 10ms, Maximum = 21ms, Average = 12ms

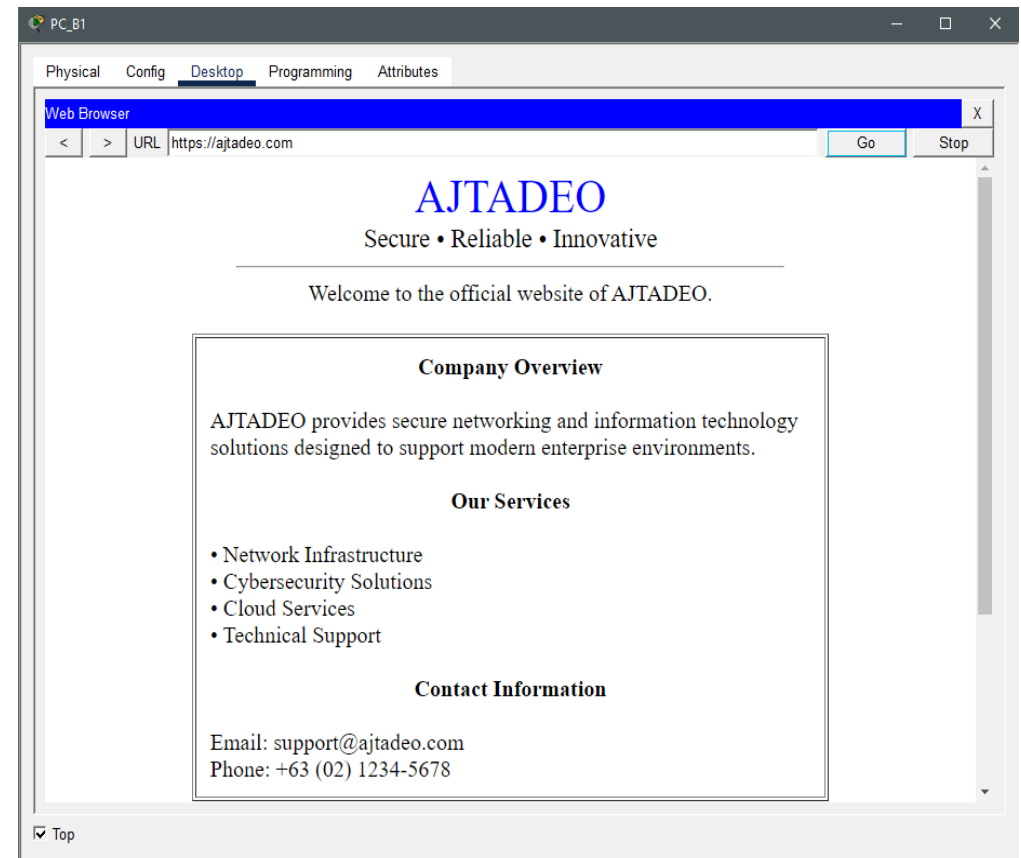
C:\>ping ajtadeo.com

Pinging 203.0.113.201 with 32 bytes of data:

Reply from 203.0.113.201: bytes=32 time=24ms TTL=118
Reply from 203.0.113.201: bytes=32 time=10ms TTL=118
Reply from 203.0.113.201: bytes=32 time=10ms TTL=118
Reply from 203.0.113.201: bytes=32 time=10ms TTL=118

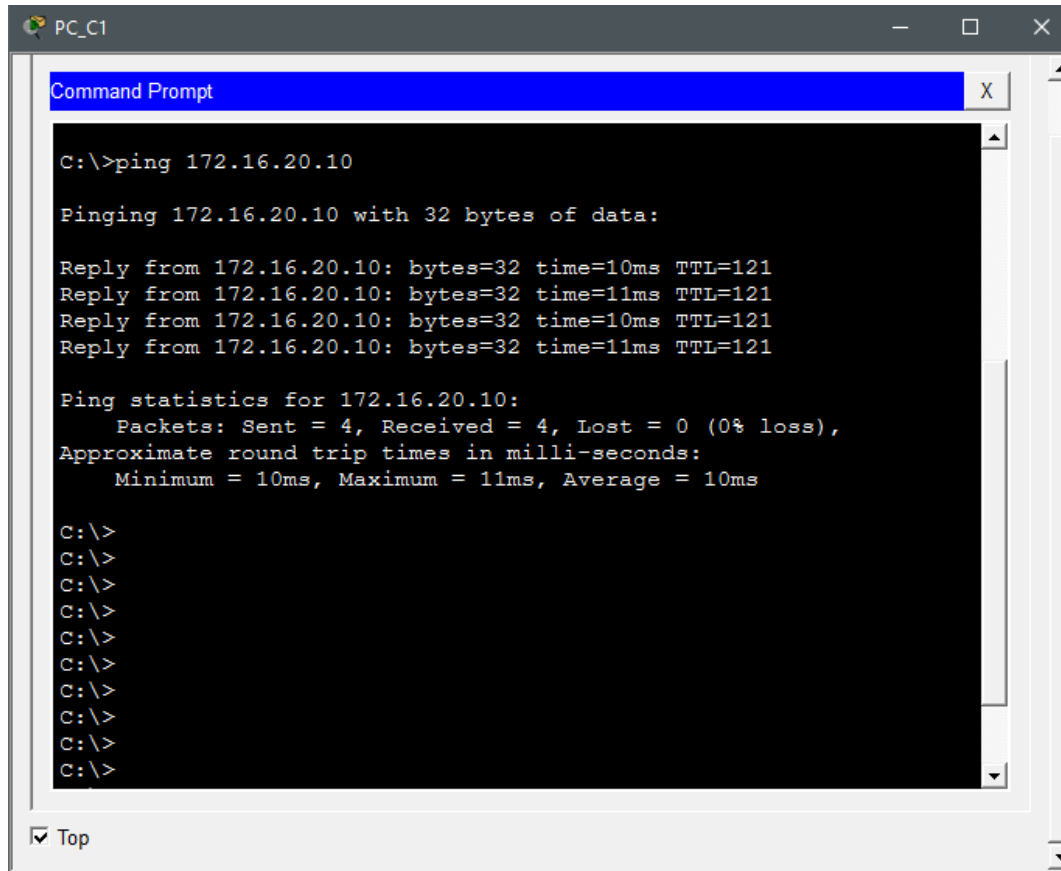
Ping statistics for 203.0.113.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 24ms, Average = 13ms

C:\>
C:\>
C:\>
C:\>
C:\>
```



*Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.*

# Verification – STORAGE (FTP)



```
PC_C1
Command Prompt

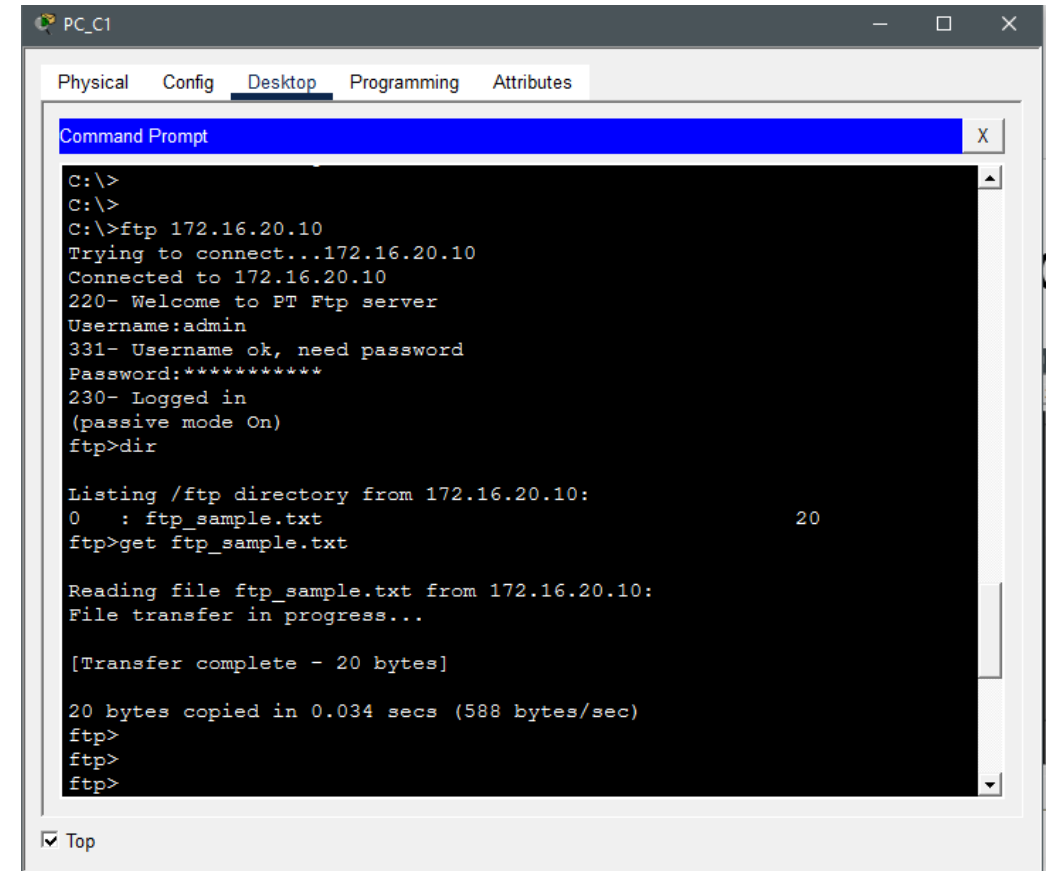
C:\>ping 172.16.20.10

Pinging 172.16.20.10 with 32 bytes of data:

Reply from 172.16.20.10: bytes=32 time=10ms TTL=121
Reply from 172.16.20.10: bytes=32 time=11ms TTL=121
Reply from 172.16.20.10: bytes=32 time=10ms TTL=121
Reply from 172.16.20.10: bytes=32 time=11ms TTL=121

Ping statistics for 172.16.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 11ms, Average = 10ms

C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
```



```
PC_C1
Physical  Config  Desktop  Programming  Attributes

Command Prompt

C:\>
C:\>
C:\>ftp 172.16.20.10
Trying to connect...172.16.20.10
Connected to 172.16.20.10
220- Welcome to PT Ftp server
Username:admin
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 172.16.20.10:
0      : ftp_sample.txt                                20
ftp>get ftp_sample.txt

Reading file ftp_sample.txt from 172.16.20.10:
File transfer in progress...

[Transfer complete - 20 bytes]

20 bytes copied in 0.034 secs (588 bytes/sec)
ftp>
ftp>
ftp>
```

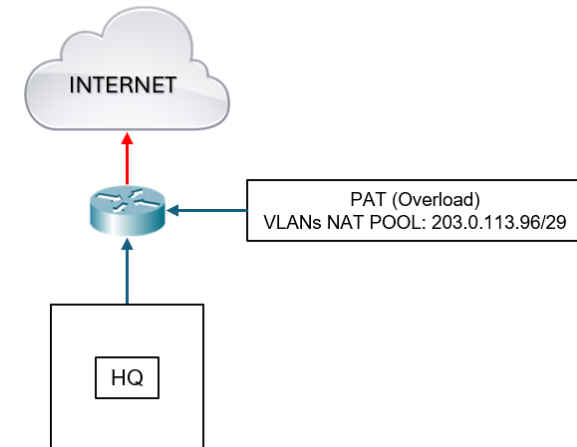
*Note: Occasional timeouts on initial ping attempts are expected due to ARP resolution and do not indicate connectivity issues.*

# Network Address Translation

| CONFIGURED ON EDGE_HQ1      |     |                             |                                                   |                                |                   |
|-----------------------------|-----|-----------------------------|---------------------------------------------------|--------------------------------|-------------------|
| NAT Type                    | ACL | Private IP (Inside Local)   | Public IP (Inside Global)                         | Inside Interface               | Outside Interface |
| PAT (Overload) via NAT Pool | 10  | All VLANs except Management | 203.0.113.96/29<br>(203.0.113.97 - 203.0.113.102) | Port-Channel 1<br>(G0/0, G0/1) | G0/1/0            |
|                             |     | All VLANs except Management |                                                   |                                | G0/2/0            |

## NAT Verification

```
EDGE_HQ1# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 203.0.113.97:1    192.168.30.11:1   8.8.8.8:1          8.8.8.8:1
icmp 203.0.113.97:2    192.168.30.11:2   8.8.8.8:2          8.8.8.8:2
icmp 203.0.113.97:3    192.168.30.11:3   8.8.8.8:3          8.8.8.8:3
icmp 203.0.113.97:4    192.168.30.11:4   8.8.8.8:4          8.8.8.8:4
icmp 203.0.113.97:5    192.168.30.11:5   8.8.8.8:5          8.8.8.8:5
icmp 203.0.113.97:6    192.168.30.11:6   8.8.8.8:6          8.8.8.8:6
icmp 203.0.113.97:7    192.168.30.11:7   8.8.8.8:7          8.8.8.8:7
icmp 203.0.113.97:8    192.168.30.11:8   8.8.8.8:8          8.8.8.8:8
```

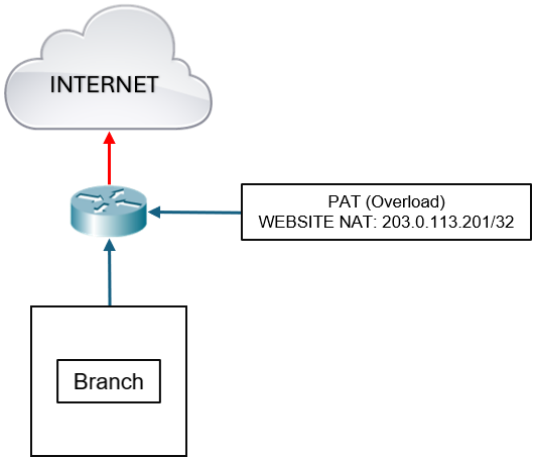


# Network Address Translation

| CONFIGURED ON EDGE_BR1              |         |                           |                                            |                                |                   |
|-------------------------------------|---------|---------------------------|--------------------------------------------|--------------------------------|-------------------|
| NAT Type                            | Service | Private IP (Inside Local) | Public IP (Inside Global)                  | Inside Interface               | Outside Interface |
| PAT (Overload)<br>via Static NAT    | WEBSITE | 172.16.10.8               | 203.0.113.201 /32<br>(Publicly Accessible) | Port-Channel 1<br>(G0/0, G0/1) | G0/1/0            |
|                                     |         |                           |                                            |                                | G0/2/0            |
| NON-NAT<br>(STRICTLY Internal Only) | PORTAL  | 172.16.20.8               | X                                          | X                              | X                 |
|                                     | FTP     | 172.16.20.10              |                                            |                                |                   |

## NAT Verification

```
EDGE_BR1# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 203.0.113.201:2    172.16.10.8:2     8.8.8.8:2          8.8.8.8:2
icmp 203.0.113.201:3    172.16.10.8:3     8.8.8.8:3          8.8.8.8:3
icmp 203.0.113.201:4    172.16.10.8:4     8.8.8.8:4          8.8.8.8:4
icmp 203.0.113.201:5    172.16.10.8:5     8.8.8.8:5          8.8.8.8:5
icmp 203.0.113.201:6    172.16.10.8:6     8.8.8.8:6          8.8.8.8:6
icmp 203.0.113.201:7    172.16.10.8:7     8.8.8.8:7          8.8.8.8:7
icmp 203.0.113.201:8    172.16.10.8:8     8.8.8.8:8          8.8.8.8:8
--- 203.0.113.201      172.16.10.8      ---                ---
```





# SECURITY - ASA Firewall Configuration

| Firewall | Inside Interface(s) | Outside Interface | Security Levels     |
|----------|---------------------|-------------------|---------------------|
| ASA-HQ1  | 10.20.10.2/30       | 10.20.10.5/30     | 100 (IN) / 0 (OUT)  |
| ASA-HQ2  | 10.20.20.2/30       | 10.20.20.5/30     | 100 (IN) / 50 (OUT) |
| ASA-BR1  | 10.30.10.2/30       | 10.30.10.5/30     | 100 (IN) / 0 (OUT)  |
|          | 10.30.10.10/30      |                   |                     |
| ASA-BR2  | 10.30.20.2/30       | 10.30.20.5/30     | 100 (IN) / 50 (OUT) |
|          | 10.30.20.10/30      |                   |                     |

| SECURITY LEVEL LEGEND |                                           |
|-----------------------|-------------------------------------------|
| Security Level        | Purpose                                   |
| 100                   | Fully Trusted (Internal Network)          |
| 50                    | Semi-Trusted (Site-to-Site Communication) |
| 0                     | Untrusted (Internet Connectivity)         |

# SECURITY – ACLs Enforcement Architecture

| Device                   | Security Function                                     |
|--------------------------|-------------------------------------------------------|
| EDGE_HQ1                 | Internet edge filtering and return-traffic validation |
| ASA-HQ1                  | Headquarters Internet perimeter protection            |
| EDGE_HQ2                 | GRE/IPsec tunnel protocol protection                  |
| ASA-HQ2                  | Controls branch-to-HQ access through tunnels          |
| Core / Distribution ACLs | Enforce Management VLAN isolation                     |
| EDGE_BR1                 | Public website traffic filtering                      |
| ASA-BR1                  | DMZ protection for the Website server                 |
| EDGE_BR2                 | Branch VPN tunnel protection                          |
| ASA-BR2                  | Protects Portal and Storage servers                   |

# ACL Security Policy Overview

## **Internal Segmentation**

- User VLANs are isolated from Management VLANs.
- Management VLAN can access all enterprise networks.
- Headquarters users can access internal branch services through GRE/IPsec tunnels.

## **Branch Service Access**

- Portal (HTTPS) and Storage (FTP) are accessible from HQ through GRE/IPsec tunnels. Management VLAN may access all branch networks and services for administration and troubleshooting.
- HQ networks may initiate communication toward branch resources.
- Branch networks and servers cannot initiate communication toward HQ networks.

## **Public Website Access**

- WEBSITE is published through NAT (203.0.113.201).
- Internet users access the website using HTTPS.
- HQ's VLAN 99 (Management) may access the website using its private IP address.

## **General Security Principle**

- Explicitly permitted traffic is allowed. All other traffic is denied.

# Access Control Summary

## HQ User VLANs (10,20,30,40,110,120,130)

- ✓ Internet access (through NAT)
- ✗ Access to VLAN 98 and VLAN 99
- ✓ HTTPS to Portal (172.16.20.8)
- ✓ FTP to Storage (172.16.20.10)
- ✓ Ping Portal
- ✓ Ping Storage
- ✗ Private access to Website (172.16.10.8)

## Management VLAN (192.168.99.0/28)

- ✗ Internet access (no NAT)
- ✓ Access to all HQ VLANs
- ✓ HTTPS to Portal
- ✓ FTP to Storage
- ✓ SSH to all Network Devices
- ✓ Ping all enterprise networks
- ✓ HTTPS to Website private IP (172.16.10.8)

## Internet Users

- ✓ HTTPS to 203.0.113.201
- ✓ Ping 203.0.113.201
- ✗ Portal access
- ✗ FTP access
- ✗ Management access

## Branch VLAN 10 (Website DMZ)

- ✗ Initiate ping toward HQ
- ✗ Access Branch VLAN 20
- ✗ Access Branch VLAN 99
- ✓ Reply traffic only

## Branch VLAN 20 (Portal/Storage)

- ✗ Initiate ping toward HQ
- ✗ Access Branch VLAN 10
- ✗ Access Branch VLAN 99
- ✓ Reply traffic only

# Layer 2 Security – Port Security

## **Enabled On**

- HQ Access Switches
- Branch Access Switches

## **Configuration**

- Sticky MAC learning enabled on end-user ports
- Maximum of 1 MAC address permitted on user-facing interfaces
- Restrict violation mode configured
- Access point interfaces configured with a maximum of 30 MAC addresses
- Sticky MAC learning disabled on access point ports to accommodate multiple wireless clients

## **Purpose**

- Prevent unauthorized device connections
- Mitigate MAC flooding attacks
- Limit end-host spoofing
- Support multiple wireless clients connected through access points

# Layer 2 Security – DHCP Snooping

| HEADQUARTERS |                 |                                                       |                                                                                                                                                               |
|--------------|-----------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enabled On   | Protected VLANs | Trusted Interfaces                                    | Purpose                                                                                                                                                       |
| ACC_HQA      | 10, 99, 110     | Uplink Ports (G0/1 – G0/2)                            | <ul style="list-style-type: none"><li>• Prevent rogue DHCP Servers</li><li>• Prevent DHCP starvations attacks</li><li>• Build DHCP binding database</li></ul> |
| ACC_HQB      | 20, 99, 120     |                                                       |                                                                                                                                                               |
| ACC_HQC      | 30, 99, 130     |                                                       |                                                                                                                                                               |
| ACC_HQD      | 40, 99          | Uplink Ports (G0/1 – G0/2)<br>Server Interface (F0/1) |                                                                                                                                                               |
| ACC_HQE      | 99              | Uplink Ports (G0/1 – G0/2)                            |                                                                                                                                                               |
| BRANCH       |                 |                                                       |                                                                                                                                                               |
| Enabled On   | Protected VLANs | Trusted Interfaces                                    | Purpose                                                                                                                                                       |
| ACC_BR1      | 10, 99          | Uplink Ports (G0/1 – G0/2)                            | Prevent rogue DHCP Servers                                                                                                                                    |
| ACC_BR2      | 20, 99          |                                                       |                                                                                                                                                               |

# Layer 2 Security – Dynamic ARP Inspection

## **Enabled On**

- Same VLANs protected by DHCP Snooping

## **Trusted Interfaces**

- Uplink Ports
- DHCP Server Port

## **Validation**

- Source MAC
- Destination MAC
- IP-to-MAC Binding

## **Purpose**

- Prevent ARP spoofing and ARP poisoning attacks
- Validate IP-to-MAC bindings
- Protect host-to-host communication

# Layer 2 Security – Disabling Unused Interfaces

**Unused interfaces were administratively disabled and isolated to prevent**

- Unauthorized device connections
- Rogue switch installations
- MAC flooding attacks
- VLAN hopping attempts
- Accidental network access

**This control was implemented on the following network devices:**

- Edge Routers
- ASA Firewalls
- Core Switches
- Distribution Switches
- Access Switches

**ALL UNUSED PORTS WERE ASSIGNED TO VLAN 999 (UNUSED\_PORTS)**



# Management VLAN Configuration

| HEADQUARTERS |                 |                 |                                            |
|--------------|-----------------|-----------------|--------------------------------------------|
| Device       | Management IP   | Default Gateway | Note                                       |
| ACC_HQA      | 192.168.98.4/28 | 192.168.98.1    | SVIs were configured for remote management |
| ACC_HQB      | 192.168.98.5/28 |                 |                                            |
| ACC_HQC      | 192.168.98.6/28 |                 |                                            |
| ACC_HQD      | 192.168.99.4/28 | 192.168.99.1    |                                            |
| ACC_HQE      | 192.168.99.5/28 |                 |                                            |
| BRANCH       |                 |                 |                                            |
| Device       | Management IP   | Default Gateway | Note                                       |
| ACC_BR1      | 172.16.99.4/28  | 172.16.99.1     | SVIs were configured for remote management |
| ACC_BR2      | 172.16.99.5/28  |                 |                                            |

# SSH Configuration for Network Devices

## Devices Covered

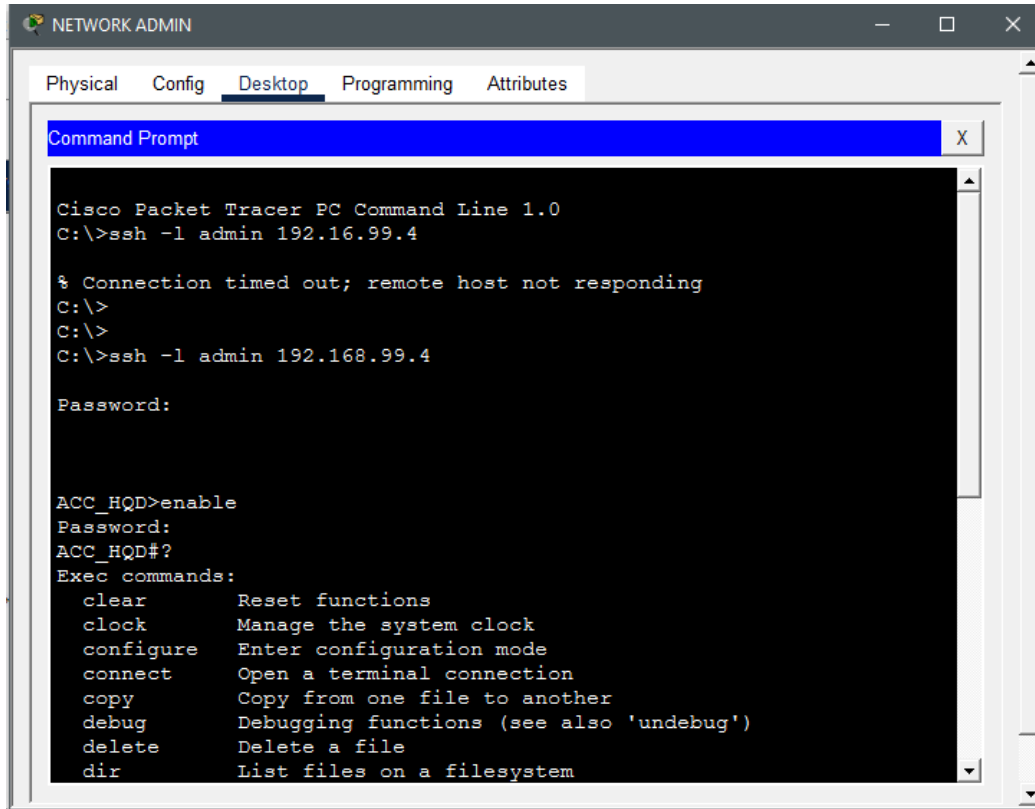
- Edge Routers
- Core Switches
- ASA Firewalls
- Distribution Switches
- Access Switches

## Security Features

- Local user authentication
- SSH version 2 enabled
- RSA encryption keys generated
- Access restricted through SSH-MGMT ACL

| Parameter           | HQ            | Branch        |
|---------------------|---------------|---------------|
| Username            | admin         | admin         |
| Enable Secret       | password123   | password123   |
| Domain Name         | hq.local      | branch.local  |
| SSH Version         | 2             | 2             |
| RSA Key Length      | 2048-bit      | 2048-bit      |
| Console Timeout     | 30 min        | 30 min        |
| Authentication      | Local         | Local         |
| Access Method       | Local / SSH   | Local / SSH   |
| Authorized SSH Host | 192.168.99.10 | 192.168.99.10 |

# Verification - SSH



NETWORK ADMIN

Physical Config **Desktop** Programming Attributes

Command Prompt

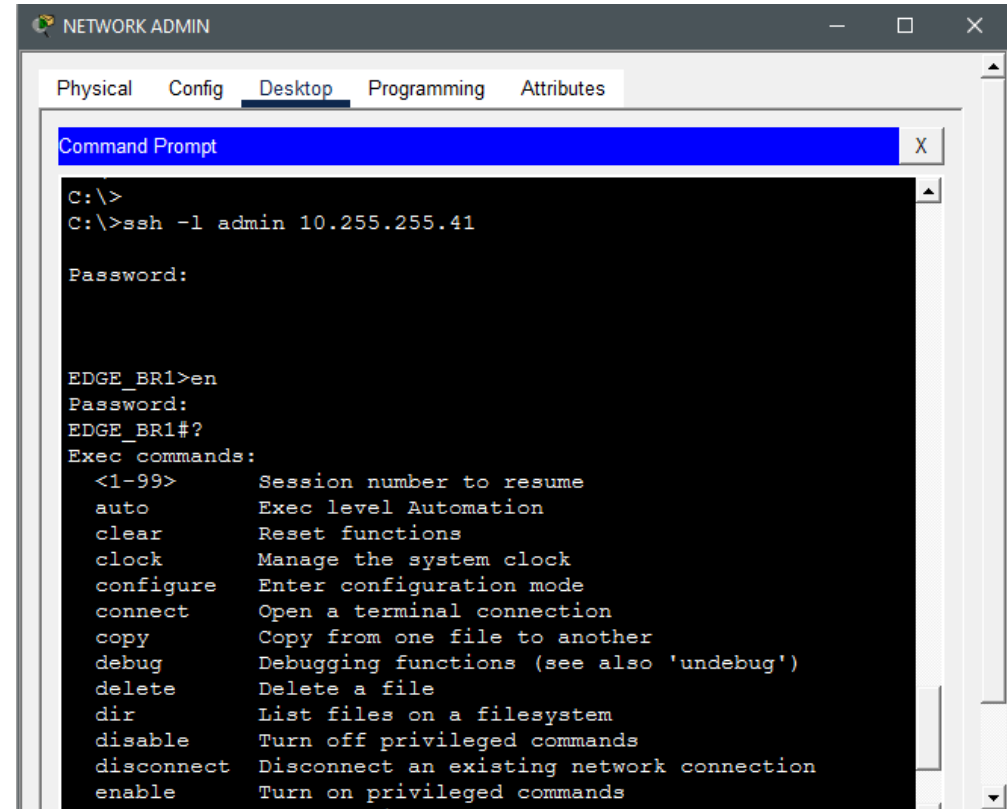
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.16.99.4

% Connection timed out; remote host not responding
C:\>
C:\>
C:\>ssh -l admin 192.168.99.4

Password:

ACC_HQD>enable
Password:
ACC_HQD#?
Exec commands:
  clear      Reset functions
  clock      Manage the system clock
  configure  Enter configuration mode
  connect    Open a terminal connection
  copy       Copy from one file to another
  debug      Debugging functions (see also 'undebug')
  delete     Delete a file
  dir        List files on a filesystem
```

SSH ACC\_HQD (HQ)



NETWORK ADMIN

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>
C:\>ssh -l admin 10.255.255.41

Password:

EDGE_BR1>en
Password:
EDGE_BR1#?
Exec commands:
  <1-99>      Session number to resume
  auto        Exec level Automation
  clear       Reset functions
  clock       Manage the system clock
  configure   Enter configuration mode
  connect     Open a terminal connection
  copy        Copy from one file to another
  debug       Debugging functions (see also 'undebug')
  delete      Delete a file
  dir         List files on a filesystem
  disable     Turn off privileged commands
  disconnect  Disconnect an existing network connection
  enable      Turn on privileged commands
```

SSH EDGE\_BR1 (BRANCH)

# Conclusion

## **Implemented Features**

- Multi-site enterprise architecture
- GRE over IPsec VPN connectivity
- OSPF and BGP dynamic routing
- NAT and PAT services
- Public Website, Portal, FTP, DHCP, DNS and WLAN services
- ASA firewall protection
- ACL-based traffic control
- Layer 2 security (Port Security, DHCP Snooping, DAI)
- Management VLAN isolation
- Secure SSH administration
- Verification of connectivity and services